

DOKUMENTASI LENGKAP & BEDAH CODING PER-FILE

WEBSITE PORTOFOLIO AKHMAD DANIL (VERSI 5.0)

React 18 | Vite 5 | Tailwind CSS | Material UI | Framer Motion | Lucide Icons

Nama Lengkap	Akhmad Danil
Program Studi	Sistem Informasi — Universitas Negeri Semarang (UNNES)
Profil Keahlian	Aspiring Data Analyst Business & Strategy Turning Data Into Decisions
Arsitektur Kode	100% Self-Contained, Bebas Database Eksternal (Supabase Removed), Clean Code
Status Kompilasi	ESLint 0 Errors 0 Warnings, Vite Build 19.9 detik (Production Ready)

BAB I: PENDAHULUAN & ARSITEKTUR TEKNOLOGI

Website portofolio ini dibangun menggunakan stack frontend modern berbasis React 18 dan Vite 5. Seluruh kode dirancang secara mandiri (self-contained) tanpa bergantung pada layanan basis data pihak ketiga seperti Supabase atau Firebase, sehingga website kebal terhadap kegagalan jaringan eksternal dan dapat memuat seketika (0 ms latency). Semua data dinamis dikonsolidasikan dalam satu berkas pusat 'src/data/portfolioData.js' sebagai Single Source of Truth.

BAB II: BEDAH KODINGAN FILE-BY-FILE SECARA MENDALAM

Di bawah ini adalah penjelasan mendalam tentang setiap berkas di dalam proyek, mulai dari file konfigurasi akar, berkas halaman (Pages), komponen UI (Components), hingga berkas data pusat.

2.1 File: index.html (Root Entry HTML)

Lokasi: index.html

Fungsi: Halaman HTML utama yang dimuat browser pertama kali. Mengatur SEO, font Poppins, Schema.org Person, dan wadah mounting React.

```
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <link rel="icon" type="image/png" href="/Photo.jpeg" />
  <title>Akhmad Danil | Aspiring Data Analyst & Business Strategy</title>
  <meta name="description" content="Portfolio Akhmad Danil, Mahasiswa Sistem Informasi
```

```

UNNES..." />
<!-- Preload Font Poppins -->
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link rel="stylesheet"
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;700&display=swap">
<!-- Schema.org JSON-LD -->
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Person",
  "name": "Akhmad Danil",
  "jobTitle": "Aspiring Data Analyst & Business Strategy"
}
</script>
</head>
<body>
  <div id="root"></div>
  <script type="module" src="/src/main.jsx"></script>
</body>
</html>

```

Penjelasan Kodingan:

- link rel='icon': Menggunakan foto formal Akhmad Danil (/Photo.jpeg) sebagai favicon di tab browser.
- Preload Google Font: Mengunduh font Poppins lebih awal sehingga tampilan tipografi langsung indah saat render pertama.
- JSON-LD Schema.org: Menyediakan metadata terstruktur yang dapat dibaca oleh crawler Google agar website mudah terindeks.
- <div id='root'></div>: Titik tumpu tempat seluruh komponen React akan dirender oleh ReactDOM.

2.2 File: src/main.jsx (React Bootstrapper)

Lokasi: src/main.jsx

Fungsi: Berkas JavaScript pembuka yang memicu inisialisasi aplikasi React 18.

```

import React from "react"
import ReactDOM from "react-dom/client"
import App from "./App.jsx"
import "./index.css"

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
)

```

Penjelasan Kodingan:

- ReactDOM.createRoot: Menggunakan arsitektur Concurrent Mode di React 18 untuk rendering yang lebih responsif.
- <React.StrictMode>: Membantu mendeteksi efek samping (side effects) atau fungsi usang selama fase pengembangan.
- import './index.css': Menyuntikkan konfigurasi Tailwind CSS ke seluruh pohon komponen.

2.3 File: src/App.jsx (Main Application & Layout Assembly)

Lokasi: src/App.jsx

Fungsi: Mengatur routing, manajemen animasi splash screen pembuka, dan lazy loading komponen.

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import React, { useState, lazy, Suspense } from "react";
import { HelmetProvider } from "react-helmet-async";
import "./index.css";
import Navbar from "./components/Navbar";
import Home from "./Pages/Home";
import About from "./Pages/About";
import Experience from "./components/Experience";
import AnimatedBackground from "./components/Background";
import { AnimatePresence } from "framer-motion";
import Footer from "./components/Footer";

// Lazy Loaded Pages
const Portofolio = lazy(() => import("./Pages/Portofolio"));
const ContactPage = lazy(() => import("./Pages/Contact"));
const WelcomeScreen = lazy(() => import("./Pages/WelcomeScreen"));
const NotFoundPage = lazy(() => import("./Pages/404"));

const LandingPage = ({ showWelcome, setShowWelcome }) => (
  <>
    <AnimatePresence mode="wait">
      {showWelcome && (
        <Suspense fallback={null}>
          <WelcomeScreen onLoadingComplete={() => setShowWelcome(false)} />
        </Suspense>
      )}
    </AnimatePresence>

    {!showWelcome && (
      <>
        <Navbar />
        <Home />
        <About />
        <Experience />
        <Suspense fallback={<div className="h-20" />>
          <Portofolio />
        } />
      </>
    )}
  </>
)
```

```

        <ContactPage />
      </Suspense>
      <Footer />
    </>
  )}
</>
);

```

Penjelasan Kodingan:

- State showWelcome: Mengontrol kemunculan WelcomeScreen. Begitu selesai animasi, bernilai false dan menampilkan halaman portofolio.
- AnimatePresence: Komponen dari framer-motion yang memastikan animasi 'exit' berjalan mulus sebelum WelcomeScreen dihapus dari DOM.
- Suspense & lazy(): Menghindari download aset sekaligus sehingga halaman dibuka sangat enteng.

2.4 File: src/data/portfolioData.js (Single Source of Truth)

Lokasi: src/data/portfolioData.js

Fungsi: Berkas pusat seluruh data website. Terdiri dari 5 ekspor objek/array yang digunakan oleh seluruh komponen.

```

export const personalData = {
  name: "Akhmad Danil",
  role: "Aspiring Data Analyst | Business & Strategy",
  quote: "Data tanpa konteks bisnis hanyalah angka; keputusan terbaik lahir dari data yang dibaca dengan tepat.",
  email: "akhmaddanil13@gmail.com",
  phone: "082138102393",
  whatsapp: "https://wa.me/6282138102393",
  linkedin: "https://www.linkedin.com/in/akhmad-danil-53430a304/",
  github: "https://github.com/ydbeksnbwk",
  typingWords: ["Aspiring Data Analyst", "Business & Strategy", "Turning Data Into Decisions", "Information Systems UNNES"]
};

export const skillsData = {
  dataAnalytics: [
    { name: "SQL (Query, Business Analysis)", level: "Advanced", icon: "database" },
    { name: "Excel (PivotTable, Data Cleaning, Dashboard)", level: "Advanced", icon: "table" },
    { name: "Python (Pandas, Data Preprocessing)", level: "Intermediate", icon: "code" },
    { name: "Machine Learning (K-Means, SVM, Random Forest)", level: "Intermediate", icon: "cpu" },
    { name: "Data Visualization", level: "Advanced", icon: "bar-chart-2" },
    { name: "Kaggle Dataset Exploration", level: "Intermediate", icon: "search" }
  ],

```

```

    businessStrategy: [
      { name: "Business Case Analysis", level: "Advanced", icon: "briefcase" },
      { name: "Market Sizing (TAM-SAM-SOM)", level: "Advanced", icon: "pie-chart" },
      { name: "Market Research & Validation", level: "Advanced", icon: "check-circle" },
      { name: "Financial Projection (Break-even, ROI)", level: "Intermediate", icon:
"trending-up" },
      { name: "Growth Strategy Planning", level: "Advanced", icon: "target" },
      { name: "Project Management", level: "Advanced", icon: "calendar" }
    ],
    tools: [
      { name: "Excel / Google Sheets", category: "Spreadsheets", icon: "sheet" },
      { name: "Google Looker Studio", category: "BI & Visualization", icon: "bar-chart-2" },
      { name: "MySQL", category: "Database", icon: "database" },
      { name: "Python (Jupyter / Colab)", category: "Programming", icon: "terminal" },
      { name: "Kaggle", category: "Platform", icon: "globe" }
    ]
  };

export const projectsData = [
  {
    id: 1,
    Title: "kmeans.predict(TBC) – Klasterisasi Penyakit TBC",
    Description: "Web tool interaktif dan riset Data Mining menggunakan algoritma K-
Means...",
    Img: "/kmeans_tbc.png",
    Tags: ["Data Mining", "K-Means (K=3)", "Clustering TBC", "Live Web Tool"],
    Link: "https://klompok11.kitaro.store/"
  },
  {
    id: 2,
    Title: "Customer Behavior & Loyalty Dashboard – SuryaHub",
    Description: "Dashboard Business Intelligence interaktif di Google Looker Studio...",
    Img: "/suryahub_dashboard.png",
    Tags: ["Looker Studio", "Data Visualization", "E-commerce BI", "Live Dashboard"],
    Link: "https://datastudio.google.com/reporting/982a9811-2582-4192-827d-0b453f2517ed"
  },
  {
    id: 3,
    Title: "Analisis PivotTable – Studi Kasus \"SuryaMart\"",
    Description: "Analisis data retail transaksi pelanggan menggunakan PivotTable
Spreadsheet...",
    Img: "/suryamart_excel.png",
    Tags: ["Excel / Sheets", "PivotTable", "Customer Analytics", "Live Spreadsheet"],
    Link: "https://docs.google.com/spreadsheets/d/e/2PACX-
1vRZsWj0DXxE3g1Fywnu1xWbZ75wMP2wM5CsEI0DhD76RRejV1R7Cq3DYp0v0lgfCg/pubhtml#gid=2113725272"
  }
];

```

Penjelasan Kodingan:

- Semua data live (TBC Web Tool, Looker Studio, SuryaMart Spreadsheet) terkoneksi langsung ke tautan publik asli.
- Menambahkan Google Looker Studio ke dalam skillsData.tools untuk mencerminkan portofolio visualisasi BI terbaru.

2.5 File: src/Pages/Home.jsx (Hero Section & Typewriter Effect)

Lokasi: src/Pages/Home.jsx

Fungsi: Bagian utama layar pembuka. Memuat headline 'Akhmad Danil', tagline, efek ketik dinamis, tombol CTA, dan ikon sosial.

```
const StatusBadge = memo(() => (
  <div className="inline-block animate-float lg:mx-0">
    <div className="relative group">
      <div className="absolute -inset-0.5 bg-gradient-to-r from-[#6366f1] to-[#a855f7]
rounded-full blur opacity-30"></div>
      <div className="relative px-3 sm:px-4 py-2 rounded-full bg-black/40 backdrop-blur-xl
border border-white/10">
        <span className="bg-gradient-to-r from-[#6366f1] to-[#a855f7] text-transparent bg-
clip-text text-sm font-medium flex items-center">
          <Sparkles className="w-4 h-4 mr-2 text-blue-400" />
          Turning Data Into Decisions
        </span>
      </div>
    </div>
  </div>
</div>
));

// Typewriter Effect Hook
const handleTyping = useCallback(() => {
  if (isTyping) {
    if (charIndex < WORDS[wordIndex].length) {
      setText(prev => prev + WORDS[wordIndex][charIndex]);
      setCharIndex(prev => prev + 1);
    } else {
      setTimeout(() => setIsTyping(false), PAUSE_DURATION);
    }
  } else {
    if (charIndex > 0) {
      setText(prev => prev.slice(0, -1));
      setCharIndex(prev => prev - 1);
    } else {
      setWordIndex(prev => (prev + 1) % WORDS.length);
      setIsTyping(true);
    }
  }
}, [charIndex, isTyping, wordIndex]);
```

Penjelasan Kodingan:

- `React.memo` pada `StatusBadge`: Mencegah komponen ini di-render ulang setiap kali state teks typewriter berubah setiap 90 milidetik.
- `useCallback` pada `handleTyping`: Mengunci referensi fungsi di memori dan hanya memperbaruinya saat `charIndex` atau `isTyping` berubah.
- `CTAButton`: Tombol navigasi 'Lihat Project' (scroll halus ke `#Portofolio`) dan 'Hubungi Saya' (scroll halus ke `#Contact`).

2.6 File: src/Pages/About.jsx (Profil, Metrik & 3 Pilar Keahlian)

Lokasi: src/Pages/About.jsx

Fungsi: Menampilkan foto profil berjas formal hitam, bio diri, kutipan visi, 4 kartu statistik capaian, dan kartu keahlian.

```
const ProfileImage = memo(() => (
  <div className="flex justify-center lg:justify-end items-center p-0">
    <div className="relative group">
      { /* 3 Lapisan Gradient Blur */ }
      <div className="absolute -inset-6 opacity-[25%] z-0 hidden sm:block">
        <div className="absolute inset-0 bg-gradient-to-r from-violet-600 via-indigo-500
to-purple-600 rounded-full blur-2xl animate-spin-slower" />
        <div className="absolute inset-0 bg-gradient-to-l from-fuchsia-500 via-rose-500 to-
pink-600 rounded-full blur-2xl animate-pulse-slow opacity-50" />
      </div>
      <div className="w-64 h-64 sm:w-80 sm:h-80 rounded-full overflow-hidden shadow-
[0_0_40px_rgba(120,119,198,0.3)]">
        
      </div>
    </div>
  </div>
));
```

Penjelasan Kodingan:

- Aksen Neon Blur: Menggunakan kombinasi gradien violet, indigo, dan purple dengan animasi rotasi lambat (animate-spin-slower) di balik foto formal.
- useMemo pada statsData: Menghindari alokasi ulang array statistik (5+ Projects, 14+ Sertifikasi, Top 10 BCC, Finalist Majuon) saat re-render.
- SkillCard: Merender daftar keahlian dengan badge level (Advanced / Intermediate) dan ikon CheckCircle2.

2.7 File: src/Pages/Portofolio.jsx (Tabs Proyek, Sertifikat & Tech Stack)

Lokasi: src/Pages/Portofolio.jsx

Fungsi: Menyediakan 3 tab interaktif dengan AppBar Material UI dan SwipeableViews.

```
// Fitur See More / See Less Sertifikat
const [showAllCertificates, setShowAllCertificates] = useState(false);
const isMobile = window.innerWidth < 768;
const initialCertificates = isMobile ? 4 : 6;

const displayedCertificates = showAllCertificates
  ? certificatesData
  : certificatesData.slice(0, initialCertificates);

// Grid 12 Tech Stack Simetris
const techStacks = [
  { icon: Terminal, language: "Python" },
  { icon: Database, language: "SQL & MySQL" },
  { icon: FileSpreadsheet, language: "Excel / Sheets" },
  { icon: BarChart2, language: "Looker Studio" },
  { icon: Cpu, language: "Machine Learning" },
  { icon: TrendingUp, language: "Market Sizing" },
  { icon: PieChart, language: "Financial Projection" },
  { icon: "javascript.svg", language: "JavaScript" },
  { icon: "reactjs.svg", language: "React" },
  { icon: "tailwind.svg", language: "Tailwind CSS" },
  { icon: "vite.svg", language: "Vite" },
  { icon: Database, language: "Data Mining" },
];
```

Penjelasan Kodingan:

- **SwipeableViews:** Memungkinkan pengguna handphone untuk menggeser tab dengan sentuhan jari (swipe gesture).
- **Pagination Toggle:** Tombol See More / See Less menjaga tampilan rapi dan tidak terlalu panjang di awal.
- **12 Item Simetris:** Angka 12 dapat dibagi habis oleh 6 (desktop), 3 (tablet), dan 2 (mobile), mencegah adanya kartu yatim di baris terakhir.

2.8 File: src/Pages/Contact.jsx (Kontak & Integrasi Email)

Lokasi: src/Pages/Contact.jsx

Fungsi: Menyediakan formulir pengiriman pesan langsung ke email tanpa butuh database.

```
const handleSubmit = async (e) => {
  e.preventDefault();
  setIsSubmitting(true);
  Swal.fire({
```

```

    title: 'Mengirim Pesan...',
    html: 'Harap tunggu selagi kami mengirim pesan Anda',
    allowOutsideClick: false,
    didOpen: () => { Swal.showLoading(); }
  });

  try {
    const formSubmitUrl = `https://formsubmit.co/${personalData.email}`;
    const submitData = new FormData();
    submitData.append('name', formData.name);
    submitData.append('email', formData.email);
    submitData.append('message', formData.message);
    submitData.append('_subject', 'Pesan Baru dari Website Portfolio Akhmad Danil');
    submitData.append('_captcha', 'false');
    submitData.append('_template', 'table');

    await axios.post(formSubmitUrl, submitData);
    Swal.fire({
      title: 'Berhasil!',
      text: 'Pesan Anda telah berhasil terkirim!',
      icon: 'success',
      confirmButtonColor: '#6366f1'
    });
    setFormData({ name: "", email: "", message: "" });
  } catch (error) {
    Swal.fire({ title: 'Gagal!', text: 'Terjadi kesalahan.', icon: 'error' });
  } finally {
    setIsSubmitting(false);
  }
};

```

Penjelasan Kodingan:

- FormSubmit Endpoint: Menggunakan Axios POST multipart/form-data langsung ke email pribadi Akhmad Danil.
- Swal Loading & Success: Menghilangkan keraguan pengunjung dengan memberikan umpan balik visual langsung.

2.9 File: `src/components/CardProject.jsx` (Smart Action Buttons)

Lokasi: `src/components/CardProject.jsx`

Fungsi: Merender setiap kartu proyek dengan deteksi otomatis link dan badge aksi.

```

const isLiveApp =
  Link &&
  (Link.includes("kitaro.store") ||
    Link.includes("spreadsheets") ||

```

```

    Link.includes("datastudio") ||
    Link.includes("lookerstudio"));

// Tombol Dinamis
{isLiveApp && (
  <a
    href={Link}
    target="_blank"
    rel="noopener noreferrer"
    className="px-3 py-1 text-[11px] font-semibold rounded-full bg-emerald-600/90 text-white flex items-center gap-1.5 shadow-md shadow-emerald-500/25"
  >
    <span className="w-1.5 h-1.5 rounded-full bg-emerald-200 animate-pulse"></span>
    <span>
      {Link.includes("spreadsheets")
        ? "Buka Spreadsheet"
        : Link.includes("datastudio") || Link.includes("lookerstudio")
        ? "Buka Dashboard"
        : "Buka Web Tool"}
    </span>
    <ExternalLink className="w-3 h-3" />
  </a>
)}

```

Penjelasan Kodingan:

- Tombol Pintar: Otomatis mendeteksi tipe link sehingga pengunjung tahu apakah mereka membuka Web Tool, Looker Studio Dashboard, atau Google Sheets.
- Tag Highlight: Tag bernilai 'Live Dashboard', 'Live Web Tool', atau 'Live Spreadsheet' diberi aksesoris border dan teks hijau emerald.

2.10 File: src/components/Certificate.jsx (Modal Lightbox Zoom)

Lokasi: src/components/Certificate.jsx

Fungsi: Menampilkan kartu sertifikat yang dapat diperbesar dengan klik untuk melihat detail tulisan.

```

const [open, setOpen] = useState(false);
const handleOpen = () => setOpen(true);
const handleClose = () => setOpen(false);

// Modal Material UI
<Modal
  open={open}
  onClose={handleClose}
  BackdropComponent={Backdrop}
  BackdropProps={{
    timeout: 300,

```

```

    sx: { backgroundColor: "rgba(0, 0, 0, 0.9)", backdropFilter: "blur(5px)" }
  }}
>
<Box sx={{ position: "relative", maxWidth: "90vw", maxHeight: "90vh" }}>
  <IconButton onClick={handleClose} sx={{ position: "absolute", right: 16, top: 16 }}>
    <CloseIcon sx={{ fontSize: 24 }} />
  </IconButton>
  <img src={ImgSertif} alt="Certificate Full View" className="max-h-[90vh] object-
contain" />
</Box>
</Modal>

```

Penjelasan Kodingan:

- Backdrop Blur: Memberikan latar belakang gelap transparan 90% saat memperbesar sertifikat.
- Link Verifikasi: Tautan '✓ Verifikasi Online →' langsung membuka bukti keaslian sertifikat di Dicoding.

2.11 File: `src/components/Experience.jsx` (Timeline Rekam Jejak)

Lokasi: `src/components/Experience.jsx`

Fungsi: Menyajikan garis waktu kompetisi dan bootcamp dengan foto panggung kompetisi.

```

const getExperienceIcon = (type) => {
  switch (type) {
    case "National Competition": return Trophy;
    case "Startup Incubation": return Rocket;
    case "Leadership & Business Strategy": return Building2;
    default: return GraduationCap;
  }
};

// Tata letak zig-zag bergantian kiri dan kanan di desktop
className={`relative flex flex-col md:flex-row items-center ${
  isEven ? "md:flex-row-reverse" : ""
}`}

```

Penjelasan Kodingan:

- Pemilihan Ikon Dinamis: Fungsi `getExperienceIcon` memilih ikon Trophy untuk lomba nasional dan Rocket untuk inkubasi startup.
- Dokumentasi Foto Panggung: Menampilkan foto asli Akhmad Danil berjas almamater kuning di panggung BCC Pegadaian dan foto tim Majuon 2025.

2.12 File: src/components/Navbar.jsx (Sticky Header & Scroll Spy)

Lokasi: src/components/Navbar.jsx

Fungsi: Navigasi atas melayang yang secara otomatis mendeteksi section layar yang sedang aktif.

```
const NAV_ITEMS = [
  { href: "#Home", label: "Home" },
  { href: "#About", label: "About & Skill" },
  { href: "#Experience", label: "Pengalaman" },
  { href: "#Portofolio", label: "Project" },
  { href: "#Contact", label: "Kontak" },
];

// Active Scroll Spy Logic
const currentPosition = window.pageY;
const active = sections.find(section =>
  currentPosition >= section.offset &&
  currentPosition < section.offset + section.height
);
if (active) setActiveSection(active.id);
```

Penjelasan Kodingan:

- NAV_ITEMS di Luar Komponen: Mencegah alokasi ulang array pada setiap render, menjaga performa scroll tetap stabil.
- Smooth Scrolling: Saat link diklik, halaman bergulir halus ke section tujuan dengan offset 100px untuk kompensasi tinggi navbar.

2.13 File: src/components/Background.jsx (Parallax Ambient Glow)

Lokasi: src/components/Background.jsx

Fungsi: Animasi lingkaran cahaya gradien di latar belakang yang bergeser mengikuti pergerakan scroll.

```
const handleScroll = () => {
  const newScroll = window.pageYOffset;
  blobRefs.current.forEach((blob, index) => {
    const initialPos = INITIAL_POSITIONS[index];
    const xOffset = Math.sin(newScroll / 100 + index * 0.5) * 340;
    const yOffset = Math.cos(newScroll / 100 + index * 0.5) * 40;
    blob.style.transform = `translate(${initialPos.x + xOffset}px, ${initialPos.y + yOffset}px)`;
    blob.style.transition = "transform 1.4s ease-out";
  });
  requestId = requestAnimationFrame(handleScroll);
};
```

Penjelasan Kodingan:

- `requestAnimationFrame`: Mengoptimalkan siklus animasi pergerakan bola cahaya agar disinkronkan dengan refresh rate layar monitor (60-144 Hz).

2.14 Berkas-Berkas Pendukung Lainnya

- **`src/Pages/WelcomeScreen.jsx`**: Splash screen pembuka dengan typewriter 'akhmaddanil.portfolio' dan animasi exit Framer Motion.
- **`src/Pages/404.jsx`**: Halaman error elegan bertema dark mode (#030014) dengan nomor 404 berpendar dan tombol kembali.
- **`src/components/SocialLinks.jsx`**: Kartu grid 4 media sosial (LinkedIn, WhatsApp, GitHub, Email) dengan efek hover neon bercahaya.
- **`src/components/Footer.jsx`**: Komponen footer sederhana yang menghitung tahun berjalan otomatis (`new Date().getFullYear()`).
- **`src/components/TechStackIcon.jsx`**: Merender kartu teknologi yang dapat menerima komponen ikon Lucide maupun URL gambar SVG.
- **`src/index.css`**: Konfigurasi scrollbar kustom bercahaya ungu-indigo dan keyframes animasi kilau (`.animate-shine`).
- **`package.json`**: Daftar dependensi yang telah dirampingkan, membuang 19 paket sampah tidak terpakai.
- **`eslint.config.js`**: Konfigurasi linter modern yang memastikan 0 error dan 0 warning.

BAB III: PANDUAN PENGOPERASIAN & DEPLOYMENT

```
# 1. Menjalankan di Komputer Lokal
npm run dev
# Buka di browser: http://localhost:5173 atau http://localhost:5174

# 2. Pengecekan Kualitas Kode (Linting)
npm run lint
# Hasil: 0 problems (0 errors, 0 warnings)

# 3. Kompilasi Produksi (Production Build)
npm run build
# Hasil: Siap dalam ~19 detik di folder 'dist/'
```



PANDUAN DEPLOYMENT: Untuk mendeploy ke Vercel secara gratis: Push kode ke GitHub -> Buka vercel.com -> Pilih repository Anda -> Klik 'Deploy'. Website langsung aktif dengan alamat HTTPS resmi.